Intelligent Employee Recommendation System

1. Context and Objectives

1.1 Context

Companies must effectively manage their human resources to assign the right people to the right activities (trainings, missions, certifications, audits, projects…). Manual selection is often long and imprecise.
This system aims to automate the selection process using AI and Natural Language Processing (NLP), taking into account skills, experience, and context.

1.2 Objectives

- Identify the employees most suited for or most in need of an activity.
- Consider skill levels and their evolution.
- Adapt selection according to context: priority to low, medium, or expert levels.
- Manage the number of available seats for each activity.
- Provide an analytical dashboard to visualize global competencies and gaps.
- Implement an intelligent module capable of understanding the description of an activity and matching it with employee profiles (job description + skills).
- Ensure traceability and post-activity skill evaluation to improve recommendations.

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2. System Modules

2.1 User Management

- Create, modify, and delete accounts.
- Manage roles and permissions: HR, managers, employees.
- Secure authentication (password, OAuth, SSO).

2.2 Employee Management

- Create and update employee profiles.
- Information: department, job position, experience, activity history.
- Participation history and post-skill evaluation.

2.3 Skill Management

- Definition of skills according to three types:
    - **Knowledge (Savoir):** theoretical knowledge
    - **Know-how (Savoir-faire):** technical/practical skills
    - **Soft skills (Savoir-être):** behavioral qualities
- Assign skills to employees with levels: low, medium, high, expert.
- Track skill evolution with dynamic scores.

2.4 Activity Management

- Define activities: training, certification, project, mission, audit.
- Define required skills and desired level (by type: knowledge, know-how, soft skills).
- Number of available seats or roles.
- Detailed description, location, date, duration.
- Prioritization based on context (upskilling, expertise, development for low/medium/high levels).

2.5 Dynamic Scoring Module

- Calculate a numeric score per skill for each employee.
- Global score for the activity based on skill weighting.
- Automatic update after participation.

2.6 Optimization and Contextual Prioritization Module

- Weight skills based on importance for the activity.
- Selection based on context: low, medium, or expert profiles.
- Manage ties or conflicts.

2.7 AI/NLP Recommendation Module

- Text analysis of activity description:
  - Extract key skills, objectives, and constraints.
  - Classify activity: technical, management, transversal.
- Matching with employee profiles:
  - Compare activity vectors ↔ employee vectors (job description + skills).
  - Combined scoring: skills, experience, progression.
  - Context-based prioritization (low/medium/expert).
- Continuous improvement:
  - Automatic retraining using manager feedback.
  - Dynamic adjustment of weights based on real effectiveness.

2.8 Post-Activity Evaluation Module

- Update skill scores after participation.
- Track activity effectiveness to improve future recommendations.

2.9 Analytical and Predictive Dashboard

- Visualize global skills by department, team, or project.
- Identify gaps and suggest activities.
- Statistics on skill progression and coverage.

2.10 Requests and Notifications Management

- Create and track activity requests by managers.
- Notifications to selected employees and managers/HR.

2.11 History and Tracking

- Complete history of recommendations and decisions.
- Traceability for internal audits.

2.12 Advanced Search and Filtering

- Search by skills, level, department, activity history.
- Scenario simulation to test different selection criteria.

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3. Recommendation Module Process

1. Input: activity description, number of seats, objectives, constraints.
2. NLP: extraction of key skills and context.
3. Transformation into semantic vectors.
4. Calculation of dynamic employee scores (by skill type).
5. Application of prioritization rules based on context.
6. Multi-criteria optimization: skills + progression.
7. Output: list of recommended employees with detailed scores.
8. Score update and feedback for continuous improvement.

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4. Detailed Matching

- Job description: match responsibilities and tasks.
- Skill sheet: match with knowledge, know-how, and soft skills.
- Dynamic scores: weighted based on:
    - Skill level
    - Recent progression
    - Context priority (develop low, consolidate medium, exploit expert)
- AI/NLP optimization:
    - Embeddings for semantic similarity
    - Supervised learning using real history
    - Conflict and tie management

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5. Deliverables

Deliverable	Description	Week
Requirements document	Validated document	S1
UML diagrams and mockups	Architecture, use cases, UI	S3
Operational backend	Data and scoring management	S5–S7
AI/NLP module	Extraction, matching, recommendation	S9
Complete frontend	Dashboard, user/activity management	S11
Tests and integration	Unit, functional, integration tests	S13
Technical & user documentation	Complete guide + final delivery	S14